Analyst Agent for Indian Public Data with Calibrated Decision Models

Milestone 1: Problem Definition and Literature Review (revised)

DS & AI Lab Project · Generative AI Track · Group 4 · 27 October 2026

This report describes a proposed investigation. It makes no claim that the system has met any target.

1. Summary

- **Problem.** Indian public statistics are open but hard to use correctly. A language model can return an executable, fluent analysis that uses the wrong indicator, year, population group or unit, and a non-expert cannot detect this.
- **System.** An agent over a curated catalogue of 8–12 tables. A hosted LLM writes read-only DuckDB queries and the narration. Rules handle units, place names and periods. Decision models make the control decisions and verify each answer. The agent shows its query, source and data version, and declines when the catalogue cannot answer.
- **Main research question.** Does a calibrated decision-and-verification layer improve end-to-end answer correctness and appropriate abstention over an LLM workflow with the same deterministic checks, at acceptable latency and cost?
- **Evidence.** One frozen 200-question test set, paired comparisons with confidence intervals, one-at-a-time ablations, and a failure taxonomy that separates data limitations from system errors.

2. Problem Definition and Scope

Obstacles for a non-expert

- **Discovery:** one indicator is published across data.gov.in, MoSPI, RBI, IMD, the agriculture ministry, NFHS and the Census. Handled by an admin-curated catalogue; outside the agent's claim.
- **Representation:** years as columns, footnote and total rows, mixed units (lakh, crore, per cent), changing place names (Orissa and Odisha; Andhra Pradesh before and after 2014).
- **Domain semantics:** usual versus current weekly labour status; July–June agricultural year versus April–March financial year; survey rounds that are not always comparable.
- **Trust:** successful execution does not show that the selected data answers the question asked.

Example. "Compare female labour force participation in Tamil Nadu and Bihar since 2018. Is the gap closing?" needs the right PLFS table, one worker-status definition and age group, and consistent survey years. A query over all ages instead of 15 and above runs without error, returns a plausible number and is wrong.

Problem statement. Build and evaluate an agent that gives a non-expert a correct, explained and verifiable answer to a question answerable from a curated catalogue of Indian public tables, in about 30 seconds. The agent states, with a calibrated confidence, how far the answer can be trusted. It declines or asks for clarification, naming the missing coverage, when the catalogue cannot support the question as asked.

In scope

- *Core (26 Nov):* fixed LangGraph workflow; hosted-LLM query writing; place, time and unit resolvers; read-only DuckDB over versioned Parquet; decision layer (Kev for tool selection and domain behind gates, Jev for coverage, verification and all other decisions); retrieval over verified examples and guidance

cards; evidence panel; decline and data-request log; streaming UI; evaluation suite; Docker Compose deployment with tracing.
- *Should (10 Dec):* admin portal; drift detection; human-approved retraining procedure.
- *Stretch:* Kev for clarification and intent; a further Kev decision; fine-tuned retrieval model; Kev-4B comparison; Kev as verifier; maps and PDF export.

Out of scope: training a query-writing model; free-form code execution; PDF table extraction; forecasting; real-time feeds; unapproved datasets; cross-portal discovery by the agent; voice input and non-English answers.

3. Stakeholders

- **Journalists and fact-checkers (primary):** citable comparisons on deadline and checks of quoted figures. Receive source, data version, executed query and stated confidence.
- **Students and researchers:** trends and plain definitions without knowing each survey's conventions.
- **NGO staff and local administrators:** district indicators without SQL or cleaning.
- **Admin team (operator):** control over which data and model changes reach users, through an approval portal, review queue and ranked list of unmet data requests.
- **Data publishers:** the data-request log shows which data users seek and cannot find.

An unsupported figure in a news story or programme plan does more harm than a withheld answer, but an unnecessary refusal also denies access to public data. Both are measured.

4. Research Questions and Contribution

Primary question

- **RQ1.** Does a decision-model control layer with calibrated verification improve end-to-end correctness and appropriate abstention compared with an LLM workflow that uses the same deterministic query and data checks?

Secondary questions

- **RQ2.** Which failure types (section 9) are reduced by Indian-data convention handling, catalogue-aware coverage checks and claim verification?
- **RQ3.** Can a fine-tuned Kev-0.8B take tool selection and domain decisions within a pre-declared margin of Jev, with fewer LLM calls, lower latency or lower cost?
- **RQ4.** Does the confidence score predict correctness of the complete answer, not only of one subtask?
- **RQ5.** How do coverage and error rate trade off as the abstention threshold changes?

Implementation objectives (not research claims): the agent and tools, evidence panel and interface, onboarding and admin path, deployment.

Intended contribution

- A benchmark of questions over selected Indian public tables, including unanswerable and ambiguous cases.
- A failure taxonomy separating data limitations from system errors.
- An empirical comparison of deterministic checks, LLM control, decision-model control and calibrated verification.
- An evidence-first prototype that states what its confidence means.

These are intended outcomes. If the decision layer does not improve on the simpler baseline, the negative result is reported and the simpler design recommended. The Kev fine-tune, its hyperparameter runs and the calibration fits are run and reported whatever the outcome.

5. Literature Review

Numbers in brackets refer to section 14.

5.1 Existing solutions

Solution	Strength	Limitation for the intended users
MoSPI eSankhyiki MCP server [1][2][3]	Official, open-source (MIT) access layer for any MCP-capable assistant; 31 datasets listed in its README (including PLFS, CPI, WPI, NAS, NFHS, RBI); four tools called in order (list datasets, list indicators, get metadata, get data)	README describes filter values and API parameters only: no definitions, unit handling, comparability checks, evidence trail or confidence. Use of the data is left to the calling assistant. Closest existing solution; a data source here
General chat assistants with code execution	Flexible analysis of an uploaded file	No catalogue of Indian data; survey definitions applied only if stated; no evidence trail or confidence
Warehouse and BI copilots: Snowflake Cortex Analyst [4], Databricks Genie [5]	Natural-language questions over governed tables; Snowflake self-reports 90%+ accuracy on an internal benchmark	Need a curated warehouse and a data-team-written semantic model; no Indian conventions
data.gov.in [6]	357,955 resources in 12,531 catalogues (22 October 2026)	Search and download only; no unit, period or comparability checks
Decision models: Jev [7]; Kev [8][9]; Laya [10]	Typed probabilities for routing and verification in one forward pass	Self-reported figures. Kev-0.8B scores 0.586 against 0.757 for Jev on Kev's own benchmark and below chance zero-shot on tool routing [9]; fine-tuning narrows the gap (Laya 0.362 to 0.766 [10])

5.2 What the research establishes and leaves open

- **Grounding improves numerical facts; coverage and interpretation remain the bottlenecks.** DataGemma grounds an LLM in Data Commons statistics [11][12]. With retrieval, cited statistics are 98.6–98.9% accurate, but inferences from them are correct in 71.9–76.4% of cases, and statistical answers appear for 24–29% of queries. Of the unanswered queries, 37.5–43.1% lacked the dataset and 30.6–40.3% failed through an incorrect data query [11]. *Consequence for this project:* missing data and failed retrieval are labelled separately, and conclusions are checked apart from raw values.
- **Executable queries are not correct answers.** Spider and BIRD made execution accuracy the standard measure [13][14]. The best BIRD model at publication reached 40.08% against 92.96% for humans [14]; later systems reached 55.9% [15] and 71.10% in a high-compute preprint [16]. DA-Code and InfiAgent-DABench score code-written analyses against reference answers [17][18]. Execution accuracy does not detect a query that runs but answers a different question, and these benchmarks use databases with consistent definitions. Official access has also improved (MoSPI MCP server), but choosing the indicator, filter and survey round remains with the assistant. *Consequence:* evaluation checks data selection and intent alignment. Query writing is bought from a hosted LLM and no query model is trained.
- **Abstention is studied as model uncertainty or schema infeasibility.** Selective classification trades coverage for risk [19]; a survey organises LLM abstention [20]; EHRSQL and TrustSQL add questions the schema or SQL cannot express [21][22]. None addresses partial coverage: a table at the wrong geography level, period, population or definition. This is the common case in Indian survey data

and the one most likely to yield a substitute statistic. *Consequence:* the test set includes partial-coverage questions and records why each question is unanswerable.

- **Confidence is meaningful only for a defined event on the target distribution.** Networks are often miscalibrated and temperature scaling corrects this in-distribution [23]. Language models partly generalise self-assessment, with poorer calibration on new tasks [24]. Open decision models report calibration on their own benchmarks [8]. No reviewed system defines what an analytic assistant's confidence refers to. *Consequence:* the confidence event is defined explicitly (section 8.2) and calibrated on human labels from this domain.
- **Learned verification and routing help in specific settings.** Trained verifiers improve mathematical reasoning [25][26]. LLM judges agree with humans over 80% of the time but show position, verbosity and self-enhancement bias [27][28]. Cascades cut cost at matched quality on answer generation [29][30]; tool-calling remains error-prone [31]. None tests routing or verification for agent control over statistical tables. *Consequence:* the control layer is compared with a strong deterministic baseline.

Synthesis. Each technique is useful alone. None ensures a correct answer, and errors compound: a wrong semantic mapping gives a valid query over the wrong data; a verifier that checks only arithmetic approves it; a confidence score calibrated to that subtask reports high confidence. The project therefore evaluates the complete pipeline and records the stage at which each failure arises.

5.3 Benchmarks, baselines and metrics

- **Reported in existing work:** execution or answer accuracy (Spider, BIRD, DA-Code, InfiAgent-DABench, DataGemma); abstention on schema-infeasible questions (EHRSQL, TrustSQL); decision accuracy, F1, AUROC, Brier score and ECE (When2Call, Kev, Laya); cost at matched quality (FrugalGPT, RouteLLM).
- **Not reported:** a displayed confidence tied to a defined end-user event; partial coverage.
- **Baselines adopted:** an LLM taking every decision, the usual agent design [32]; the same LLM with deterministic checks; Jev as the reference for open decision models.

6. Research Gaps

The gap is not that no system combines every feature. It is that the following problems cause harm and are not addressed by current approaches.

- **G1. Semantic mismatch despite successful execution.**
  - Consequence: a journalist publishes an official-looking figure for the wrong year, population or definition.
  - Not addressed: execution accuracy cannot detect it; benchmarks use consistent definitions.
  - Test: data-selection and intent labels on every question; end-to-end accuracy as the primary metric.
- **G2. Partial coverage answered with a substitute statistic.**
  - Consequence: a state figure reported as a district figure, or an adjacent indicator used for the one requested.
  - Not addressed: abstention benchmarks cover schema infeasibility only.
  - Test: partial-coverage test questions; false-answer rate; abstention precision and recall.
- **G3. Missing data and failed retrieval conflated.**
  - Consequence: users are told data is absent when it exists; publishers receive false data requests.
  - Not addressed: DataGemma reports both causes at similar rates under one coverage label.
  - Test: separate failure labels (section 9); oracle-table runs.

- **G4. Confidence without a defined target.**
  - Consequence: a high score can reflect correct arithmetic over the wrong table.
  - Not addressed: calibration work fixes the score, not the event it describes.
  - Test: calibration against end-to-end correctness; ECE per failure type.
- **G5. Benefit of learned control assumed, not measured.**
  - Consequence: added complexity, cost and a further failure mode with no demonstrated gain.
  - Not addressed: routing gains are shown for answer generation, not control decisions.
  - Test: comparison with an LLM-plus-checks baseline; ablations.

Latency and cost are engineering objectives (O5, O6), not research gaps.

7. Approach

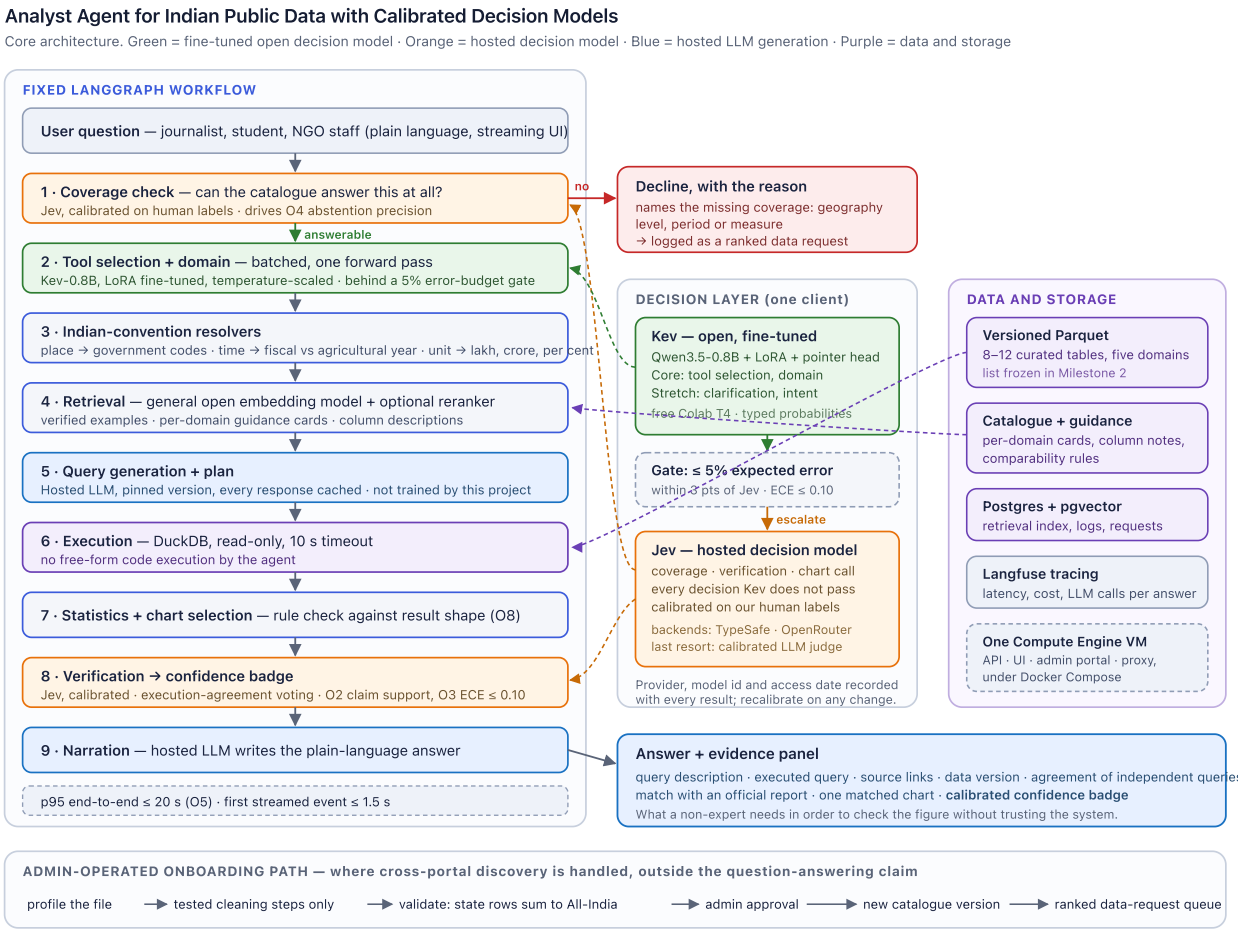


Figure 1. Core architecture. Discovery sits in the admin-operated onboarding path, outside the question-answering claim.

A fixed LangGraph workflow runs typed tools in a set order. Each task goes to the simplest mechanism that can perform it.

Deterministic (rules, code, lookup)

- SQL is well formed, read-only, runs within 10 s; schema and types match (sqlglot, DuckDB).
- Units, canonical place names, fiscal and agricultural years.
- Arithmetic, grouping and ranking in DuckDB; post-checks for empty results, value ranges and state rows summing to the All-India row.

- Numbers quoted in the answer matched against the result table.
- Explicit coverage: requested geography level or year range absent from catalogue metadata.

Learned (needs semantic judgement)

- Partial or semantic coverage: does this table measure what was asked? Jev, calibrated on human labels.
- Tool selection and domain: Kev-0.8B, fine-tuned, behind a gate; uncertain calls escalate to Jev.
- Whether the result answers the question as intended, and whether textual claims are supported: Jev verifier, calibrated. Its input includes the selected table and filters; otherwise a wrong selection cannot be detected.
- Text generation (queries, plans, narration, clarifying questions): hosted LLM.

Why a decision model rather than another LLM call. It returns typed probabilities in one forward pass, several decisions can be batched, and its probabilities can be calibrated directly. For tool selection and domain, the expected gain is cost and latency (RQ3). The reliability claim rests on coverage and verification (RQ1). Kev takes a decision only if its accuracy on human labels is within a pre-declared margin of Jev's (initially 3 points) and its calibrated ECE is at most 0.10; otherwise the decision stays on Jev.

Systems compared for RQ1 (knowledge, retrieval and prompts held fixed)

- **A.** LLM takes every decision, with all deterministic checks and its own stated confidence.
- **B.** A, with Kev (escalating to Jev) for tool selection and domain.
- **C.** B, with calibrated coverage, calibrated verification and the confidence badge.

If B or C does not improve on A, the simpler design is preferred.

8. Correctness, Trust and Confidence

8.1 Levels of correctness

- **L1 Execution validity.** The query is safe, well formed and runs. *Measured:* automatically.
- **L2 Data selection.** Table, indicator, geography, period, population and definition match the request (failure: total employment used for female labour force participation). *Measured:* gold tables and intent slots.
- **L3 Numerical correctness.** The result matches the reference under the intended filters and units. *Measured:* execution accuracy against the reference result.
- **L4 Evidence support.** Every material claim is supported by the result and cited source (failure: the narrative claims a rise the table does not show). *Measured:* claim-support rate; numbers automatically, text by human labels.
- **L5 Interpretation.** Comparisons, rankings and trends follow from the evidence; units and caveats are stated. *Measured:* human rubric on a sample.
- **L6 Intent alignment.** The response answers the question asked at the requested scope, or asks for clarification (failure: state figures returned for a district question). *Measured:* human label against intent slots.
- **L7 End-to-end correctness.** L1–L6 hold and no material unsupported claim remains. The primary metric.

For an unanswerable question, the correct outcome is a decline that names the missing coverage. A clarifying question is correct only if the question is ambiguous. A refusal of an answerable question is a failure.

8.2 What the confidence score means

- **Definition.** The badge estimates the probability that the complete answer is correct at L7, given that the agent answered. Successful execution alone does not make a label positive.
- **Construction.** One calibrator (logistic or isotonic regression, chosen on the development set) over the verifier score, coverage probability, routing confidences and deterministic check outcomes, fitted on a human-labelled calibration split.
- **Evaluation.** ECE, Brier score, reliability plots with bin counts, bootstrap intervals and a risk–coverage curve on a held-out set. ECE is also reported per failure type, which shows where the score is blind, for example to a wrong table.
- **Internal scores.** Per-decision confidences of Kev and Jev control escalation, are evaluated separately (E3) and are never shown as answer confidence.
- **Display rule.** No numerical badge is shown until calibration has been tested. If calibration is inadequate, the interface shows a qualitative status or the agent abstains.

9. Failure Taxonomy: Data Problems and System Errors

Data-side

- **D1 Catalogue gap.** Indicator, geography, period or granularity absent from the catalogue. Remedy: decline with reason, log a data request, consider onboarding.
- **D2 Metadata gap.** Data exists but definitions, units or survey notes are missing. Remedy: improve metadata and guidance cards.
- **D3 Source limitation.** Survey rounds not comparable, or the published table is inconsistent. Remedy: answer with a stated caveat, or decline.

System-side

- **S1 Wrong data selection.** Suitable data exists; another table is used. Remedy: retrieval, catalogue descriptions.
- **S2 Query error.** Wrong filter, join, aggregation or period. Remedy: verified examples, prompts, query constraints.
- **S3 Execution error.** Query fails, times out or breaks a safety rule. Remedy: validation, bounded retries.
- **S4 Calculation error.** Correct result transformed wrongly downstream. Remedy: deterministic computation, unit tests.
- **S5 Unsupported inference.** Narrative adds a claim the evidence does not support. Remedy: claim checks.
- **S6 Verifier false acceptance.** Incorrect answer approved. Remedy: verifier labels, thresholds, independent checks.
- **S7 Over-answering.** Unanswerable question answered, often with a substitute statistic. Remedy: coverage decision.
- **S8 False abstention or false absence.** Agent declines, or reports data as missing, although a valid path exists. Remedy: separate coverage search from execution; review thresholds.
- **S9 Wrong clarification decision.** Clarification asked when unneeded, or not asked when needed. Remedy: labels and thresholds.

How the evaluation tells them apart

1. Each test and development question carries labels for answerability (answerable, partial, unanswerable, ambiguous), the reason if unanswerable, gold tables, reference query and result, and intent slots (indicator, geography, period, population, unit).

2. Every run is traced by stage (catalogue lookup, retrieval, routing, query, execution, checks, verifier, narration). A failure is attributed to the first stage that departs from the gold labels: a wrong table is S1; a right table with a wrong result is S2 or S4; a right result with a wrong narrative is S5.
3. Oracle runs supply the gold table, then the reference query, to separate selection errors from generation errors.
4. A table of answerability against behaviour (answered or declined) separates S7 from S8, and D1 from S8.
5. Two annotators label failure classes on a shared subset; Cohen's kappa is reported.

10. Evaluation Plan

10.1 Objectives and targets

Targets are initial. The accuracy target is fixed after the first baseline is measured. All targets are reported whether or not met.

#	Objective	Metric	Target
O1	Correct answers	End-to-end correctness (L7) on answerable questions; EX and data-selection accuracy as diagnostics	Exceed baseline A; paired difference with 95% interval
O2	Faithful explanation	Claim-support rate (L4)	At least 0.95
O3	Calibrated trust	ECE of the badge against L7; Brier score; reliability plots	At most 0.10, with bootstrap interval
O4	Appropriate decline	Abstention precision and recall; false-answer rate on unanswerable questions	Precision at least 0.85
O5	Responsiveness	Time to first streamed event; p95 end-to-end latency	At most 1.5 s; at most 20 s
O6	Efficiency	Cost per answered question; LLM calls per answer	Lower than baseline A
O7	Measurable improvement	Change in O1–O6 across versions and ablations	Every version reported
O8	Appropriate visuals	Chart type matches the rule table for the result shape; two raters, 100-answer sample	At least 0.90

10.2 Experiments

- **E1 Version ladder.** v0 (LLM takes every decision) → v1 domain knowledge and retrieval → v2 verified examples → v3 prompt optimisation (DSPy [33], GEPA [34]) → v4 Kev for tool selection and domain → v5 calibrated coverage, verification and badge → v6 recalibration and prompt re-optimisation after drift (TextGrad [35]). The ladder is cumulative, so v6 exceeding v0 is expected and serves as a check; the evidence for RQ1 and RQ2 comes from E2 and E5.
- **E2 Control comparison (RQ1).** Systems A, B and C at the v3 configuration, plus a general-purpose assistant connected to the MoSPI MCP server as an external reference on a 50-question subset.
- **E3 Decision gates (RQ3).** Per decision: zero-shot and fine-tuned Kev-0.8B, Jev, the LLM, and a Kev-to-Jev cascade. Learning-rate, LoRA-rank, epoch and data-size runs always reported. Metrics: accuracy, F1, AUROC, Brier, ECE, share handled by Kev at a 5% error budget, cost, latency.
- **E4 Verification and confidence (RQ4).** Jev with and without calibration, an LLM judge, execution-agreement voting. Metrics: AUROC, Brier, ECE per failure type, risk at fixed coverage.
- **E5 Ablations (RQ2, RQ5).** Remove domain guidance, retrieval, Kev, verifier, calibration and coverage gate one at a time; oracle-table and oracle-query runs.
- **E6 Drift simulation.** Injected burst of new-topic or new-format questions; time to detect, recovery, no regression on the frozen set.

10.3 Evaluation rules

- With 200 questions, a score near 50% has a 95% interval of about ±7 points. Systems are compared on the same questions with paired bootstrap or McNemar tests; the paired difference and interval are

reported, not a pass/fail threshold.

- Prompts, thresholds, calibrators and cascade bands are tuned on development and calibration sets only. The frozen test set is run only to produce reported results.
- Decision models, verifiers and judges are scored against human labels. Accuracy and ECE come from one canonical backend; provider, model identifier and access date are recorded, and any backend change triggers recalibration.

11. Data, Feasibility and Risks

Sources. data.gov.in (GODL-India; attribution required, terms checked per dataset [36]); MoSPI PLFS, CPI and WPI via eSankhyiki or its MCP server (published tables, not microdata; the PLFS design changed from January 2025, so series before and after need comparability notes [2][37]); IMD, Ministry of Agriculture (DES), RBI (DBIE) and NFHS (IIPS) tables under each agency's terms. The 8–12 tables are profiled and frozen in Milestone 2.

Planned data

- At least 1,000 execution-verified question and reference-query pairs.
- 1,000–2,000 routing examples per Kev decision, labelled by rules or a teacher model and checked by hand. These labels measure agreement, so every gate is scored on human labels.
- 150–200 human-labelled cases per decision, including partial-coverage variants.
- 150–200 human-labelled end-to-end answers for confidence calibration.
- Development set of 150; frozen test set of 200 (at least 20 unanswerable, some ambiguous), split by template and table and never used for training or tuning.

Risks and mitigation

- **Kev fails its gate** (Kev-0.8B is below chance zero-shot on tool routing [9]): the decision stays on Jev and the result is reported; domain is the safer second decision.
- **Synthetic questions easier than real ones:** hand-written questions in development and test sets; template and table splits; leakage checks.
- **Verifier approves wrong answers:** deterministic checks first; false-acceptance rate and ECE per failure type reported.
- **Over-abstention:** false-abstention rate and risk–coverage curves reported; answerable edge cases in the test set.
- **Jev access, version or price changes** [7]: one decision client with interchangeable backends; a calibrated LLM judge as last resort.
- **GPU limits** (T4 has no bf16; the Google Cloud trial has no GPUs [38]): week-1 smoke test, per-epoch checkpoints, rented L4 as backup.

The week-1 feasibility checks (hosted LLM on about 20 questions over 2 tables; Kev-0.8B against Jev on 200 routing examples; one-hour T4 smoke test) have not been run. Results will be reported in Milestone 2.

12. Resources, Team and Next Milestone

- **Infrastructure.** One Compute Engine VM under Docker Compose (API and agent, UI, admin portal, Postgres with pgvector, reverse proxy); Langfuse Cloud free plan for tracing. Kev-0.8B is fine-tuned on a free Colab T4 and served on a GPU endpoint for demonstrations only; evaluation runs on a MacBook or CPU.
- **Budget.** Estimated US$65–170 to 21 December, of which US$41–83 is eligible for the US$300 Google Cloud credit [38]. Team estimates.

- **Team.** T1 Data and Tools: Shaoni Samanta (21f1003157). T2 Agent: Shivanshu Gupta (22f3002985). T3 Model: Sathya Prakash (21f1006001). T4 Ops and Evaluation: Ashmit Ghose (24f1001827). Tracks rotate between milestones; contributions are recorded in `WORKLOG.md`.
- **Milestone 2 (29 October).** Profile and freeze the table list; build the first cleaned tables and catalogue entries with unit, place and period checks; define the labelling schema of section 9; produce the first question and reference pairs with development and test splits; report the week-1 results.

13. Conclusion

Grounding, executable queries, routing, verification, calibration and abstention each address part of trust, and none ensures that a response answers the question asked. The project succeeds if it establishes which components improve end-to-end reliability and which add complexity without benefit. A well-supported negative result is an acceptable outcome.

14. References

All links were opened between 21 and 27 October 2026. Items marked *self-reported* are vendor or project documents whose performance figures have not been independently verified.

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